

PS-06 — your data model

LensGuide — Visual Search & AR Companion

Kognivera Hackathon 2026 · Travel & Tourism · data model v1.1.0-rc2

The problem statement itself, the 24-hour MVP scope and the XR device requirement live in the hackathon application, on your statement's page. This document is the data you have been given to build it with: every table, every field, and what each one is for.

You have **12,289 rows across 13 tables**. 12 of them are the tables this statement is built on; the remaining 1 is a reference table the others point at, included so the database works on its own.

All of it is in the `data/` folder beside this document: as `PS-06.db` (SQLite, indexed, ready to query), as CSV, and as DDL for Postgres and SQLite.

What the data gives you

60 labelled landmark classes × 10 images with a train/eval split and a deliberate difficulty gradient, 200 menu/sign images with ground-truth text *and* a reference translation, and 900 grounded POI facts with honest confidence bands.

Watch out for this one

Train on `dataset_split='train'` and report on `'eval'`. Mixing them produces a number that means nothing and a judge will ask.

The tables this statement is built on

TABLE	ROWS	WHAT YOU USE IT FOR
<code>activities_poi</code>	900	Points of interest with real depth — cost, duration, carbon, hours and tags — because APS-09 optimises over exactly these fields.
<code>cities</code>	60	The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.
<code>countries</code>	30	ISO country reference. Every city, currency default and calling code resolves here.
<code>currencies</code>	25	carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).
<code>languages</code>	26	Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.
<code>landmark_image_set</code>	600	60 labelled classes x 10 images, split train / eval. Without inputs, PS-06's core AI feature is unbuildable.

TABLE	ROWS	WHAT YOU USE IT FOR
menu_sign_images	200	Ground-truth source text plus a reference translation, so in-place translation can be scored rather than eyeballed.
place_kb	1,200	chunked, embeddable prose. If retrieval returns filler the grounding claim collapses, so this text is written to be worth retrieving.
poi_facts_kb	900	Short grounded facts — what PS-06 returns beside a recognised landmark and what APS-03 puts on an info card.
poi_media	1,800	Labelled imagery per POI — the recognition training and evaluation surface for PS-06.
poi_travel_matrix	6,418	precomputed edge costs between POIs. This is what removes the Google Maps dependency for APS-09 and makes results comparable across teams.
xr_scenes	60	One immersive scene per previewable space. PS-05 embeds it on the booking page; APS-07 loads it in WebXR. Both point at the same canonical supply row.

Reference tables, included so the database is valid

You will mostly join through these rather than think about them.

TABLE	ROWS	WHAT IT IS
categories	70	One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.

How they fit together

Open `02_DATA_MODEL_DIAGRAM.html` in a browser for the clickable version — it shows these tables and nothing else. Download it first; it will not render inside SharePoint.

Some tables point at "any bookable thing" using an `(entity_type, entity_id)` pair rather than a typed foreign key. That is deliberate: it is what lets one feature refer to a hotel, a flight, a point of interest or a package without a separate join table for each. The legal values of `entity_type` are in `data/enums.json`.

Every field, table by table

Columns marked **PK** are the primary key. **FK** shows what a column points at. Enum columns list their legal values — anything else is rejected by the conformance check.

categories

Reference & geography · 70 rows · IDs start `cat_` · reference table

One two-level taxonomy shared by POI type, package theme and expense category — so the three never drift apart.

COLUMN	TYPE	CONSTRAINTS	NOTES
category_id	text	PK	cat_ prefixed.
code	text	UNIQUE · NOT NULL	snake_case.
label	text	NOT NULL	
parent_category_id	text	FK → categories.category_id	Null for top-level; self-referencing.
applies_to	text	NOT NULL	poi
updated_at	timestampz	NOT NULL	

currencies

Reference & geography · 25 rows · IDs start `cur_`

carries the true minor-unit exponent so JPY/KWD display correctly even though storage is always DECIMAL(12,2).

COLUMN	TYPE	CONSTRAINTS	NOTES
currency_id	text	PK	cur_ prefixed.
iso4217	char(3)	UNIQUE · NOT NULL	e.g. INR.
name	text	NOT NULL	
symbol	text	NOT NULL	
minor_unit_exponent	smallint	NOT NULL	0 for JPY/KRW, 2 default, 3 for KWD/BHD.
display_locale	text	NOT NULL	BCP-47 locale used for formatting.
updated_at	timestampz	NOT NULL	

languages

Reference & geography · 26 rows · IDs start `lng_`

Rule R6: BCP-47 is the only legal way to say 'language' anywhere in the model.

COLUMN	TYPE	CONSTRAINTS	NOTES
language_id	text	PK	lng_ prefixed.
bcp47	text	UNIQUE · NOT NULL	e.g. ta, hi, en-IN — never 'Tamil'.
english_name	text	NOT NULL	
native_name	text	NOT NULL	
script	text	NOT NULL	ISO-15924, e.g. Taml, Deva, Latn.
rtl	bool	NOT NULL	Right-to-left rendering flag.
tts_supported	bool	NOT NULL	Relevant to PS-13 voice output.

COLUMN	TYPE	CONSTRAINTS	NOTES
updated_at	timestamptz	NOT NULL	

countries

Reference & geography · 30 rows · IDs start `cnt_`

ISO country reference. Every city, currency default and calling code resolves here.

COLUMN	TYPE	CONSTRAINTS	NOTES
country_id	text	PK	Canonical ID, cnt_ prefixed.
iso2	char(2)	UNIQUE · NOT NULL	ISO-3166-1 alpha-2, e.g. IN.
iso3	char(3)	UNIQUE · NOT NULL	ISO-3166-1 alpha-3, e.g. IND.
name	text	NOT NULL	English short name.
default_currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	ISO-4217 code.
calling_code	text	NOT NULL	E.164 country calling code, e.g. +91.
region	text	NOT NULL	UN sub-region grouping.
updated_at	timestamptz	NOT NULL	Rule R4: UTC, ISO-8601 with offset.

cities

Reference & geography · 60 rows · IDs start `cty_`

The geographic anchor of the whole model. 60 cities; every hotel, POI, package, advisory and weather row hangs off one.

COLUMN	TYPE	CONSTRAINTS	NOTES
city_id	text	PK	cty_ prefixed.
name	text	NOT NULL	City name.
state	text		State / province, nullable for city-states.
country_id	text	FK → <code>countries.country_id</code> · NOT NULL	
country_code	char(2)	NOT NULL	Denormalised ISO2 for convenient joins.
lat	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
lng	decimal(9,6)	NOT NULL	Rule R7: WGS-84, 6dp.
timezone	text	NOT NULL	IANA zone, e.g. Asia/Kolkata.

COLUMN	TYPE	CONSTRAINTS	NOTES
region	text	NOT NULL	Domestic region grouping, e.g. South India.
population	int		Approximate, for demand weighting.
season_profile	text	NOT NULL · one of winter, summer, monsoon, post_monsoon, spring, autumn	Dominant season at the peak travel window.
peak_months	text	NOT NULL	Comma-separated month numbers, e.g. 10,11,12.
primary_language	text	FK → languages.bcp47 · NOT NULL	Rule R6: BCP-47 tag.
description	text		One-paragraph orientation blurb, used by PS-13.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

menu_sign_images

Content, knowledge & safety · 200 rows · IDs start sgn_

Ground-truth source text plus a reference translation, so in-place translation can be scored rather than eyeballed.

COLUMN	TYPE	CONSTRAINTS	NOTES
image_id	text	PK	sgn_ prefixed.
city_id	text	FK → cities.city_id · NOT NULL	
kind	text	NOT NULL	menu
file_path	text	NOT NULL	
source_text_truth	text	NOT NULL	Ground-truth OCR text.
source_language	text	FK → languages.bcp47 · NOT NULL	
reference_translation	text	NOT NULL	Reference target text.
target_language	text	FK → languages.bcp47 · NOT NULL	
script	text	NOT NULL	ISO-15924.
difficulty	text	NOT NULL	easy

COLUMN	TYPE	CONSTRAINTS	NOTES
dataset_split	text	NOT NULL · one of train, eval	
is_safety_critical	bool	NOT NULL	APS-03 — a confidently mistranslated warning is worse than an unsure one.
updated_at	timestampz	NOT NULL	

xr_scenes

XR & immersive · 60 rows · IDs start xrs_

One immersive scene per previewable space. PS-05 embeds it on the booking page; APS-07 loads it in WebXR. Both point at the same canonical supply row.

COLUMN	TYPE	CONSTRAINTS	NOTES
scene_id	text	PK	xrs_ prefixed.
entity_type	text	NOT NULL · one of hotel, room_type, rate_plan, flight, flight_fare, poi, package, package_component, guide, transfer, event, xr_scene	hotel
entity_id	text	NOT NULL	Canonical ID of the space being previewed.
city_id	text	FK → cities.city_id · NOT NULL	
name	text	NOT NULL	
scene_type	text	NOT NULL · one of panorama_360, model_3d, photogrammetry, video_360	
default_yaw_deg	smallint	NOT NULL	Initial camera heading.
default_pitch_deg	smallint	NOT NULL	
real_scale_metres	decimal(6,2)		Longest dimension — PS-05's AR mode needs true scale.
total_bytes	int	NOT NULL	Asset budget check for the judging bar.
supports_webxr	bool	NOT NULL	
supports_ar_placement	bool	NOT NULL	
fallback_media_id	text		hotel_media or poi_media row used when XR is unsupported.

COLUMN	TYPE	CONSTRAINTS	NOTES
licence	text	NOT NULL	Synthetic or openly-licensed only.
status	text	NOT NULL · one of <code>active</code> , <code>inactive</code> , <code>archived</code> , <code>draft</code>	
updated_at	timestampz	NOT NULL	

activities_poi

Supply & catalogue · 900 rows · IDs start `poi_`

Points of interest with real depth — cost, duration, carbon, hours and tags — because APS-09 optimises over exactly these fields.

COLUMN	TYPE	CONSTRAINTS	NOTES
poi_id	text	PK	poi_ prefixed.
city_id	text	FK → <code>cities.city_id</code> · NOT NULL	
name	text	NOT NULL	
category_id	text	FK → <code>categories.category_id</code> · NOT NULL	
poi_category	text	NOT NULL · one of <code>heritage</code> , <code>nature</code> , <code>museum</code> , <code>religious</code> , <code>adventure</code> , <code>food</code> , <code>shopping</code> , <code>nightlife</code> , <code>beach</code> , <code>wildlife</code> , <code>wellness</code> , <code>viewpoint</code>	Denormalised top-level category.
lat	decimal(9,6)	NOT NULL	
lng	decimal(9,6)	NOT NULL	
typical_duration_minutes	int	NOT NULL	APS-09 constraint input.
entry_cost	decimal(12,2)	NOT NULL	0.00 where free.
currency	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
carbon_kg	decimal(8,3)	NOT NULL	On-site footprint estimate.
popularity_score	smallint	NOT NULL	0–100, drives APS-04 baseline ranking.
value_score	smallint	NOT NULL	0–100, editorial worth-the-stop rating. In APS-09 this is the value signal the <code>eval_optimizer_cases</code>

COLUMN	TYPE	CONSTRAINTS	NOTES
			reference plans were built on (highest value_score per unit of entry_cost). It is not one of the three capped objectives — cost, time and carbon are — so a higher-value plan is a legitimate way to beat the reference.
opens_at	text		Local HH:MM; null where always open.
closes_at	text		
closed_days	text		Comma-separated 0–6, 0 = Monday.
best_season	text	one of winter, summer, monsoon, post_monsoon, spring, autumn	
accessibility	text	NOT NULL	step_free
tags	text	NOT NULL	Comma-separated free tags for hybrid retrieval.
description	text	NOT NULL	Plausible prose.
has_xr_scene	bool	NOT NULL	PS-05 / APS-07.
status	text	NOT NULL · one of active, inactive, archived, draft	
updated_at	timestampz	NOT NULL	

landmark_image_set

Content, knowledge & safety · 600 rows · IDs start lmk_

60 labelled classes x 10 images, split train / eval. Without inputs, PS-06's core AI feature is unbuildable.

COLUMN	TYPE	CONSTRAINTS	NOTES
image_id	text	PK	lmk_ prefixed.
poi_id	text	FK → activities_poi.poi_id	Null for generic classes.
label_class	text	NOT NULL	The 60-way recognition label.
file_path	text	NOT NULL	
dataset_split	text	NOT NULL · one of train, eval	
width_px	int	NOT NULL	

COLUMN	TYPE	CONSTRAINTS	NOTES
height_px	int	NOT NULL	
capture_conditions	text	NOT NULL	daylight
occlusion_pct	smallint	NOT NULL	Deliberate difficulty gradient.
licence	text	NOT NULL	Synthetic or openly-licensed only — no real people.
updated_at	timestampz	NOT NULL	

place_kb

Content, knowledge & safety · 1,200 rows · IDs start kbc_

chunked, embeddable prose. If retrieval returns filler the grounding claim collapses, so this text is written to be worth retrieving.

COLUMN	TYPE	CONSTRAINTS	NOTES
chunk_id	text	PK	kbc_ prefixed.
city_id	text	FK → cities.city_id	One of city_id / poi_id is always set.
poi_id	text	FK → activities_poi.poi_id	
section	text	NOT NULL · one of history, culture, etiquette, food, transport, safety, seasonal, practical	
title	text	NOT NULL	
body	text	NOT NULL	120–260 words of plausible prose.
language	text	FK → languages.bcp47 · NOT NULL	
token_estimate	int	NOT NULL	
source_label	text	NOT NULL	Synthetic provenance label, shown in citations.
embedding_ref	text		Row key into the precomputed embeddings parquet.
seasonal_relevance	text	one of winter, summer, monsoon, post_monsoon, spring, autumn	Null where the chunk is season-agnostic.
updated_at	timestampz	NOT NULL	

poi_facts_kb

Content, knowledge & safety · 900 rows · IDs start fct_

Short grounded facts — what PS-06 returns beside a recognised landmark and what APS-03 puts on an info card.

COLUMN	TYPE	CONSTRAINTS	NOTES
fact_id	text	PK	fct_ prefixed.
poi_id	text	FK → activities_poi.poi_id · NOT NULL	
fact_type	text	NOT NULL	history
fact_text	text	NOT NULL	One or two sentences, verifiable in tone.
language	text	FK → languages.bcp47 · NOT NULL	
confidence	text	NOT NULL · one of high, medium, low	Lets PS-06/APS-03 demo honest low-confidence handling.
display_priority	smallint	NOT NULL	
embedding_ref	text		
updated_at	timestampz	NOT NULL	

poi_media

Supply & catalogue · 1,800 rows · IDs start pmd_

Labelled imagery per POI — the recognition training and evaluation surface for PS-06.

COLUMN	TYPE	CONSTRAINTS	NOTES
media_id	text	PK	pmd_ prefixed.
poi_id	text	FK → activities_poi.poi_id · NOT NULL	
file_path	text	NOT NULL	
media_role	text	NOT NULL · one of hero, room, lobby, exterior, dining, pool, bathroom, view, landmark, menu, sign, receipt	
alt_text	text	NOT NULL	
label_class	text	NOT NULL	Recognition class label.
dataset_split	text	NOT NULL · one of train, eval	F9 / train vs eval split.
width_px	int	NOT NULL	
height_px	int	NOT NULL	
capture_conditions	text		daylight

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>updated_at</code>	timestampz	NOT NULL	

`poi_travel_matrix`

Supply & catalogue · 6,418 rows · IDs start `ptm_`

precomputed edge costs between POIs. This is what removes the Google Maps dependency for APS-09 and makes results comparable across teams.

COLUMN	TYPE	CONSTRAINTS	NOTES
<code>matrix_id</code>	text	PK	ptm_ prefixed.
<code>origin_poi_id</code>	text	FK → <code>activities_poi.poi_id</code> · NOT NULL	
<code>dest_poi_id</code>	text	FK → <code>activities_poi.poi_id</code> · NOT NULL	
<code>mode</code>	text	NOT NULL · one of <code>walk</code> , <code>cycle</code> , <code>auto_rickshaw</code> , <code>cab</code> , <code>bus</code> , <code>metro</code> , <code>train</code> , <code>ferry</code> , <code>flight</code>	
<code>minutes</code>	int	NOT NULL	
<code>distance_km</code>	decimal(7,3)	NOT NULL	
<code>carbon_kg</code>	decimal(8,3)	NOT NULL	
<code>cost</code>	decimal(12,2)	NOT NULL	0.00 for walk.
<code>currency</code>	char(3)	FK → <code>currencies.iso4217</code> · NOT NULL	
<code>bearing_deg</code>	smallint	NOT NULL	APS-03 wayfinding — bearing origin → destination.
<code>updated_at</code>	timestampz	NOT NULL	

Unique together: (`origin_poi_id`, `dest_poi_id`, `mode`)

The rules that apply to these fields

#	RULE
R1	Additive only. Add columns, tables and stores freely. Never rename, drop or repurpose a field that came with the data.
R2	IDs are opaque prefixed strings — <code>htl_a91f3c</code> . Never integers, never parsed for meaning.
R3	Money is a pair: a 2-place decimal plus an ISO-4217 currency code. Never a float.
R4	Time is ISO-8601 with an offset. <code>_at</code> fields carry an offset; <code>_date</code> fields have no zone.
R5	Enums are lowercase snake_case and the legal values are in <code>data/enums.json</code> .
R6	Language is a BCP-47 tag — <code>ta</code> , not "Tamil".

#	RULE
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R7	Geography is WGS-84 to 6 decimal places, <code>lat</code> and <code>lng</code> together or not at all.
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R8	Nothing is hard-deleted. Rows carry <code>status</code> and <code>updated_at</code> .
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Add whatever you like beside these fields — new columns, new tables, your own vector store, your own services. That is the point of R1. What you must not do is rename or re-key the fields that came with the data, because that is what would stop sixteen independent builds being put together afterwards.

`data/WORKING_WITH_THE_DATA.md` has the loading instructions, including how to read money without corrupting it. `tools/validate_conformance.py` tells you in thirty seconds whether you are still conformant.